Supplementary material for *Typed Issue Specifications from App Reviews: What Scales, and a Downstream Benefit That Does Not Replicate*

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This file carries the implementation settings and the annotator instruments. It is referenced from the manuscript and is not required to follow the argument there. We separate it because an audit of the manuscript against its own text found that every instrument was named but not specified, so a reader could reconstruct the architecture and not a single number. What follows is the repair, at a level of detail the manuscript has no room for.

1 Implementation Settings

Section **??** states what each stage does and why. This appendix gives the settings. An audit of this paper against its own text found that every instrument was named but not specified, so a reader could reconstruct the architecture and not a single number. What follows is the repair. Values are as they appear in the released code; where a setting is a default we say so.

Stage 1, classification. Backbone `roberta-base` (cased, 125M), multi-label sigmoid head over seven classes, per-label cut 0.5. Fine-tuning: 3 epochs, learning rate 2×10^{-5}, batch size 16, max sequence length 256, AdamW at library defaults, seed 42. Training corpus is the relabelled corpus under a stratified per-class cap of 15,000 rows, shuffled at seed 42; class order follows Python dictionary insertion order, which Section **??** shows is load-bearing for the held-out split. HITL-1 routes a review to a human when top-class probability < 0.70 or the top-two margin < 0.15.

Stage 1, the V2 annotator. The noisy corpus labels come from `gemini-2.0-flash` at temperature 0.1, 150 max output tokens, prompted for a single primary class plus a self-reported confidence, with four written tie-break rules (device-specific faults route to `compatibility`, "slow" and "lag" to `performance`, and so on). The prompt is released in full. On any API error or parse failure the caller returns `other` with confidence 0.0, so a reader cannot distinguish a genuine `other` from a failed call; this is a defect we report rather than a design choice. V1, V3 and V4 are intermediate checkpoints kept for the record and carry no reported result.

Stage 1, the verified-anchor procedure. Because this is the contribution we defend, we state it as an algorithm. (i) The anchor is 5,230 reviews the lead author labelled by hand, drawn from the strata where the V2 labeller was least productive; 5,038 of them are `praise`, so the anchor is not a balanced sample and we do not claim it is one. (ii) Out-of-sample predicted probabilities come from `sklearn.cross_val_predict` with 5 folds over a TF-IDF plus linear pipeline. (iii) `cleanlab.filter.find_label_issues` is called with those probabilities and the noisy labels, ranked by self-confidence, and `get_label_quality_scores` supplies a per-row score. (iv) A flagged row is corrected to the anchor's label only when the anchor's confidence is ≥ 0.70 and the V2 label's probability is ≤ 0.20; otherwise the original label is kept. (v) Rows the anchor does not cover, which is 97.6% of the corpus, are corrected to cleanlab's argmax under the same probability ceiling. The

3×3 grid over $(0.65, 0.70, 0.75) \times (0.15, 0.20, 0.25)$ in Section **??** sweeps (iv). "Anchor confidence" is the annotator's recorded confidence in their own label, not a model quantity.

Stage 1, compatibility augmentation. 200 template-generated synthetic reviews (device, OS, and screen templates) plus 100 keyword-mined RRGen reviews. Section **??** reports two defects in this augmentation that a reader needs before using the class.

Stage 2. Embeddings from `all-MiniLM-L6-v2`. UMAP with `n_components` = 50, `min_dist` = 0.0, `metric` = cosine, and `n_neighbors` tuned per class (30 for `bug_report`, 25 for `feature_request`, 20 for `performance` and `usability`, 10 for `compatibility`). HDBSCAN on the reduced space with Euclidean metric and per-class `min_cluster_size` / `min_samples` of 200/20, 100/10, 60/8, 60/8, and 10/3 respectively. Aspects are spaCy noun chunks filtered against a curated keyword list, released with the code. PageRank runs undirected on the bipartite review-aspect graph at NetworkX defaults (damping 0.85). The KG-hierarchical side runs over an 8,404-review sample; the flat baseline runs over 91,368 clustered reviews from the full corpus, so the two describe different review sets (Section **??**).

Stage 3. Headline generator `claude-opus-4-7`; comparators `llama-3.3-70b-versatile` via Groq at temperature 0.2 and 1500 max tokens, and `Qwen/Qwen2.5-3B-Instruct` and `Qwen/Qwen2.5-1.5B-Instruc` decoded greedily at 600 max tokens. None of these identifiers is a dated snapshot, which is a reproducibility defect we should have avoided. The generation prompt is released in full and is byte-identical across the local comparators. Routing uses the V5 predicted class of the cluster; `praise` and `other` have no template and fall back to the bug-report form, which means the pipeline emits no meaningful specification for the corpus's largest class. The pipeline re-prompts at most twice with dimension-level feedback when the mean rubric score is below 3.0.

Stage 4. ChromaDB over 15,100 documents from five sources (10,000 RRGen pairs, 5,000 similar responses, 60 changelogs, 40 FAQ entries, and the IssueSpec injected per query rather than pre-indexed). The re-run generator is `claude-opus-4-7` in both arms. Neither arm's file stores its prompt or its retrieved context, so the manipulation is not recoverable from the artifact; Section **??** states what follows from that.

Stage 5. Five policies on distilGPT2 (82M) from 400 SFT samples, 296 KTO pairs, and 100 DPO pairs; the Lagrangian loop runs 30 steps at batch size 4 with KL coefficient 0.05, and the compliance sweep covers $\tau \in \{0.0, 0.5, 0.7, 0.9, 1.0\}$. The binding-constraint re-run uses `Qwen/Qwen2.5-1.5B-Instruct` with LoRA ($r = 8$, $\alpha = 16$, on `q,k,v,o_proj`) for 90 steps.

Instruments. Strict template-fill requires, per field: title ≥ 4 words; description ≥ 30 words; ≥ 3 reproduction steps, each ≥ 5 words and containing one of a 33-verb action list released with the code; expected and actual behaviour ≥ 8 words each; ≥ 3 acceptance criteria of ≥ 8 words; a severity in P0–P3; and an affected component of ≥ 2 words not on a 10-item generic stoplist. All comparisons are "at least". SpecCov counts substantive-token overlap with the source cluster, tokens of length ≥ 5 after stop-word filtering, with quoted-string and coverage bonuses, banded to 1–5; Section **??** withdraws it as a faithfulness measure. The five-dimension rubric and the Y/P/N purity rubric are scored by `Qwen/Qwen2.5-3B-Instruct` decoded greedily; both judge prompts are released, the judge does not see the source reviews when scoring the accuracy dimension, and a spec whose output does not parse into all five dimensions is dropped without a counter. The compliance scorer is a regular expression over roughly thirty tokens across four dimensions; Section **??** reports what it does and does not detect.

2 Annotator Guidelines

The lead author (PhD in software engineering with prior mobile-app review-mining experience) served as primary annotator across all five annotation tasks. The 490-review classification gold was additionally annotated by two further independent human raters on the identical item set, aligned by review id; all three first completed a shared 20-review calibration round, disagreements were adjudicated before the main task opened, and the released gold label is the majority vote over the three raters. All human annotation in this work is carried out by those same three people. For Stage 4, all three raters scored all 200 rows of the reported re-run independently and blind to condition; the discarded 400-row round had the lead author as primary rater with one additional rater on a 30-pair subset, and is described here only because the paper reports its withdrawal. Separately, two Qwen2.5-3B-Instruct prompts (concise role-based and chain-of-thought) served as the second and third raters in the 99-review cluster-purity Krippendorff α calibration, which is an LLM-assisted panel and not one of the human rounds. For the 5,230-record verified anchor, which covers 5,229 distinct corpus rows, annotators classified each review into one of the seven classes in our extension of the Maalej-Nabil taxonomy (`bug_report`, `feature_request`, `performance`, `usability`, `compatibility`, `praise`, `other`) with multi-label assignment permitted when a review carried multiple concerns; reviews below the "fairly confident" threshold were skipped and excluded from the anchor. For the 490-review classification gold standard, a stratified sample of 70 reviews per class was drawn from the working corpus, and the lead author re-labeled each review independently and blinded to the V2 LLM prediction; the 490 gold labels serve as the held-out evaluation set for Cohen's κ progression (Table **??**) and per-class F1 (Table **??**). For the 5-dimension IssueSpec rubric (all 100 specs; an earlier stratified $n = 28$ subsample, 6 per issue type plus 2 extras), each spec was scored on a 1–5 Likert scale across five dimensions: *completeness* (are all required template fields filled?), *accuracy* (do field values correctly reflect the source cluster?), *actionability* (would a developer know what to do?), *specificity* (does the spec name concrete entities rather than generic terms?), and *clarity* (is the spec easy to read and unambiguous?), with each dimension scored independently. For the reported Stage 4 evaluation (100 reviews $\times$ 2 conditions, the same model with and without the IssueSpec, 200 rows), all three raters rated *quality* (1–5), *specificity* (1–5), and *helpful* (Y/N) while blinded to the generation method; the superseded 400-row round used four template-composed conditions and is retained in the release for comparison. In both rounds condition labels were replaced with random A/B/C/D codes and unsealed only at aggregation, with the assignment persisted as spreadsheets with JSON master keys for audit. A second human rater independently scored a 30-pair (`no_spec`, `full`) subset for inter-rater calibration. For the Y/P/N cluster-purity rubric (initial 50-cluster audit; 99-cluster three-rater subsample), annotators read 5 representative reviews per cluster and assigned Y (all 5 share the same sub-theme within the cluster's aspect), P (3 or 4 of 5 share the sub-theme), or N (fewer than 3 share); weighted purity is $\text{purity}_w = (|Y| + 0.5|P|)/(|Y| + |P| + |N|)$, with Y contributing 1.0, P contributing 0.5, and N contributing 0. All comparative evaluations were blinded throughout the rating window and unsealed only at aggregation; reviews or specs below the annotator confidence threshold were marked as "skip" and excluded from downstream analysis.

3 Tables moved from the manuscript

These three tables are referenced from the manuscript. They were moved here to keep the submission inside the journal's length limit; every number the manuscript's argument relies on is stated in its text, and these carry the per-cell detail.

Use	Model (family)	Result
Stage 3 generate	Claude (Anthropic) / Llama-3.3-70B (Meta) / Qwen-3B (Alibaba) / Qwen-1.5B (Alibaba)	0.96 / 0.80 / 0.74 / 0.48
Stage 4 generate	*withdrawn* (see note)	*withdrawn*
5-dim rubric judge	Qwen-3B, all 100 specs	3.94/5 (28-spec subsample: 3.89)
Cross-family rubric	Claude / Llama / Qwen-3B / Qwen-1.5B	4.09 / 3.96 / 3.83 / 3.77 (§??)
Y/P/N purity judge	Qwen-3B vs lead-author	0.625 vs 0.66; $\Delta = -0.035$

Table 1: Multi-LLM evidence summary across three model families (Anthropic, Meta, Alibaba). Cross-family rank preservation in Stage 3 is the structural-bias control; LLM-as-judge calibration Δ vs the lead author bounds judge bias. *Two rows carry caveats a reader needs.* The Stage-4 row previously reported 0.61 versus 0.54 for "Claude" against Qwen-3B. We withdraw it: the file labelled Claude is `responses_rrgen_baseline.json`, which our own release notes record as the output of a deterministic template composer rather than of any language model, so the row compared a template against a generation. This is the same confound we withdrew in Experiment 2 and we had not noticed it here. The Stage-3 row is a real cross-family comparison but is not matched: Claude is scored on 100 clusters and Llama on 99, both spanning five issue types, whereas both Qwen conditions are 15 clusters of `bug_report` only. The 0.96-versus-0.48 span therefore compares a five-type mean against a single-type mean and should be read as an ordering, not as a magnitude.

4 Full SpecCov validation

This is the analysis Section 5 of the manuscript summarises.

An extractive-coverage score is only useful if it agrees with what a person calls faithful. Three human raters independently scored 60 specs, 20 from each of three generators, shuffled and unlabelled, on a 1–5 faithfulness scale, and separately listed every claim the source reviews do not support. One qualification first: one rater completed the corrected sheet, which shows exactly the evidence SpecCov scores against, while the other two completed an earlier draft showing a subset of it, so their flags are biased toward over-flagging and we lead with the rater who saw the full evidence. The metric fails all three checks on either reading. *It does not rank specs the way humans do:* Spearman ρ between SpecCov and human faithfulness is +0.045 ($p = 0.73$), +0.015 ($p = 0.91$), and +0.226 ($p = 0.08$) for the three raters, and +0.147 ($p = 0.26$) against their mean, none significant. *It does not detect the claims humans call unsupported:* one rater marked 45 of the 60 specs as containing at least one unsupported detail and 15 as clean, and SpecCov scores those groups at 3.64 and 3.93, a difference in the right direction but far from significant (Mann-Whitney $p = 0.34$), where a metric that detected hallucination would separate them. *It orders the generators backwards:* it ranks the taxonomy condition (4.20 on this subsample) above the human reference (3.45) where the humans rank them the other way, reference 4.10 and taxonomy 3.50, because templated specs reuse source vocabulary densely, which the score rewards, while a human writer condenses and paraphrases, which it penalizes even when nothing is invented.

We therefore withdraw SpecCov as a faithfulness measure. Lexical grounding is not a usable proxy for faithfulness on this task, and the automatic 320-spec rubric that appeared to support

dimension	taxonomy, $n = 100$	taxonomy, $n = 28$	lead-author ref, $n = 20$
completeness	**3.72**	3.79	3.20
accuracy	**4.43**	4.39	4.10
actionability	**3.86**	3.71	2.85
specificity	**4.15**	4.07	4.05
clarity	**3.52**	3.50	3.05
overall mean	**3.94**	3.89	3.45
sd across specs	0.59	–	0.41

Table 2: Experiment 1 five-dimension rubric (Qwen-judge, 1–5 scale) on the *complete* spec set: all 100 taxonomy specs and all 20 lead-author reference specs, scored in one run so the two columns are directly comparable. The middle column repeats the earlier stratified 28-spec subsample under the same judge and rubric; the gap between it and the full set is +0.05, which is the subsample error. The LLM-as-judge protocol follows the calibration-against-reference convention of [**?**, **?**, **?**].

it is not independent evidence, since its faithfulness dimension is computed by the same overlap procedure. Entailment-based measurement [**?**, **?**, **?**] is the direction we would now take. We release the scorer, the 60-spec human ratings, and this analysis, because a documented negative result on a plausible-looking metric is more useful than a metric that was never tested.

Cross-LLM replication (bias control). Llama-3.3-70B reaches 0.80 template-fill, Qwen2.5-3B 0.74, and Qwen2.5-1.5B 0.48 under the same criteria, and the templated > free-form > raw ordering holds across all four (Table S1 in the supplementary material, with the two caveats its caption records, one a withdrawn row). On two per-class structural checks Llama-3.3-70B attains complete template compliance on the stratified sample, all 29 bug specs carrying three or more action-verb reproduction steps (vs. Claude's 73%) and all 30 feature specs the As-a/I-want/So-that triple (vs. 97%); these measure structural compliance, not content correctness. What the bias control shows is cross-family rank preservation: the IssueSpec contract transports across families rather than depending on one proprietary generator.

Cross-family rubric quality. Template-fill says nothing about content quality, so we ran the same Qwen2.5-3B judge and rubric over the other generators' specs on matched clusters. On the 14 clusters all four cover (bug reports only, where the two Qwen runs overlap), mean rubric score follows the template-fill order: Claude 4.09, Llama-3.3-70B 3.96, Qwen2.5-3B 3.83, Qwen2.5-1.5B 3.77. At $n = 14$, or $n = 13$ for the three pairs involving Qwen2.5-1.5B, none of the six pairwise differences reaches significance (Wilcoxon p 0.18 to 0.75), and the adjacent gaps are the size of the run-to-run generation noise of Section **??**, so the ordering is consistent with the template-fill result rather than independent confirmation of it. A wider Claude-versus-Llama comparison on 28 clusters spanning all five issue types gives 4.01 against 3.64, a nominal +0.38 at $p = 0.015$, which we previously asserted as the one cross-family quality difference we stand behind. We withdraw that assertion: it belongs to a family of seven Wilcoxon tests with the same judge and rubric, Bonferroni or Holm correction puts it at $p = 0.104$, and an exact sign test on the same 28 pairs gives $p = 0.078$. We no longer claim a cross-family quality difference at all. The rubric spread is much narrower than the (unmatched) template-fill spread, which is itself informative: the capability gap shows up mostly as whether fields get filled, with completeness separating the models (4.00 down to 3.00) and accuracy nearly flat (4.50 for three of four). One further caveat: the judge is itself Qwen2.5-3B-Instruct, so that row carries a self-preference risk the others do not.

Five-Dimension Rubric. The rubric [**?**, **?**, **?**] is operationalized with Qwen2.5-3B-Instruct as judge over the complete set: all 100 taxonomy specs and all 20 lead-author reference specs in one run (Table S2 in the supplementary material). The taxonomy condition reaches 3.94/5 (sd 0.59),

Annotator pair	sample	agreement / deviation
Lead-author vs V2 LLM	5,230	75% / **25%** mislabeled
Lead vs V2 / cleanlab / V5	489 gold	κ: 0.163 $\rightarrow$ 0.333 $\rightarrow$ **0.592**
same, vs 3-rater majority gold	489 gold	κ: 0.165 $\rightarrow$ 0.334 $\rightarrow$ **0.590**
3 human raters, Y/N verdict	490 gold	Fleiss $\kappa = \mathbf{0.968}$; 479/490 unanimous on the verdict, 476/489 on the label
3 human raters, 7-class label	489 gold	Fleiss $\kappa = \mathbf{0.979}$; $\alpha = 0.979$
V2 LLM vs V5; cleanlab vs V5	215,583	71.96% / 86.77% agree
cleanlab vs V5 (consistency, not independent)	40,291	88.66% support
V5 vs gold, held out (replay-dependent, see §**??**)	307	$\kappa = \mathbf{0.616}$
3 human raters (Stage 4 quality, discarded round)	400 rated rows	weighted κ 0.970–0.986; within-1 100%
3 human raters (Stage 4 quality, reported round)	200 rated rows	weighted κ 0.03, 0.04, 0.50
3 human raters (Stage 4 helpful, discarded round)	400 rated rows	κ 0.678–0.836; $\alpha = 0.781$
3-rater *LLM-assisted* classification check (lead + 2 Qwen prompts)	99 reviews	Krippendorff $\alpha =$ 0.285
Lead vs Qwen-judge (purity, rubric)	50 / 28	$\Delta = -0.035$ / $+0.23$

Table 3: Annotator-vs-LLM and inter-annotator deviation. The $\approx 25\%$ deviation is measured on the human-verified anchor; the κ-progression reaches moderate agreement [**?**]; the V5 endorsement row is a consistency check, not independent validation, since V5 trained on those corrections; LLM-judge calibration Δ bounds judge bias. The $\alpha = 0.285$ row is an LLM-assisted panel, corrected from a mis-implemented 0.451 as described in the text.

above the hypothesized lower bound of 3.5, and an earlier stratified 28-spec subsample gave 3.89, so subsampling cost 0.05 of a Likert point. Per issue type it ranges from 4.23 on performance down to 3.45 on usability. The reference set is a judge calibration, not a competitor: it scores 3.45, with the gap concentrated in actionability (2.85 versus 3.86), because the lead author left template-specific fields empty, writing a reference spec by hand being a different task from filling a routed template. The comparison therefore says that a human writing freely produces less tracker-ready output than a type-routed generator. It is not evidence that the specs are better than expert judgement on content, and we do not read it that way.

Stage 1 Classification Recovery. The Cohen's κ trajectory against the 490-review expert gold crosses three Landis-Koch [**?**] agreement bands (Table S3 in the supplementary material): the V2 LLM achieves $\kappa = 0.163$ (slight); after cleanlab correction [**?**] $\kappa = 0.333$ (fair); the V5 classifier retrained on the corrected labels reaches $\kappa = 0.592$ (moderate). The trajectory does not depend on any single annotator: scored against the majority-vote gold of the three-rater human panel

($n = 489$ after excluding one review left without a correction label), the same three classifiers reach $\kappa = 0.165$, 0.334, and 0.590, within 0.002 at every step. On the held-out 307-review subset (183 of the 490 entered training, mostly through the stratified cap rather than the anchor proper), V5 reaches $\kappa = 0.616$. We previously wrote that this is "slightly above the 0.592 on the full 490, ruling out a contamination artifact." Two audits make that untenable as written.

First, "held out" is not exactly true. All 100 keyword-mined `compatibility` rows in the training augmentation are verbatim copies of reviews already in the corpus, appended past index 215,583 with the label changed. Eight of them duplicate a held-out review, seven with the same label the expert assigned, and the index-based replay cannot see this because it filters on position rather than text. Dropping those eight gives $\kappa = 0.609$ on $n = 298$.

Second, and more seriously, the split is a replay rather than a recorded fact, and 0.616 is the most favourable replay. The reconstruction shuffles each class with seed 42 and iterates classes in Python dictionary insertion order. Iterating them in sorted order instead, changing nothing else, gives $\kappa = 0.594$; seeds 0, 1, 2, 7, 123 and 2024 give 0.602, 0.605, 0.617, 0.590, 0.583 and 0.606. The published figure is at the top of that range, and the direction of the inequality against 0.592 flips under most alternatives. What survives is the weaker statement: held-out κ is indistinguishable from full-set κ, across replays and after removing the duplicated rows. That still argues against memorisation, which is what the sentence was for, but it does not do so by being larger, and we should not have leaned on a direction this fragile.

The per-class recovery makes the structural collapse visible. Macro F1 against the gold rises from 0.22 (V2) through 0.38 (cleanlab-V2) to 0.65 (V5), and on the two minority classes the V2 LLM cannot see at all, F1 goes from 0.00 to 0.83 for `compatibility` and 0.00 to 0.77 for `performance`, the compatibility figure resting on the 300 augmented rows discussed below. Applied across the full corpus, V5 agrees with the cleanlab correction on 88.66% of the 40,291 corrected cases, against 71.96% for the V2 labeller and 86.77% for the cleanlab-corrected labels. V5 was trained on a corpus in which 38,985 of those corrected rows carry the corrected label, so its agreement is substantially a measure of fit to its own training signal, not an independent second opinion. We report it as a consistency check and do not claim the two signals are independent. The choice of confident-learning thresholds is not load-bearing: a 3×3 grid over $(\text{anchor\_conf}, \text{max\_llm\_prob})$ keeps V5 endorsement in an 82.7–91.0% band, with the headline $(0.70, 0.20)$ at 88.66% near the grid center.

The held-out split itself needs describing, because it is not unbiased in the way we first said. V5's training corpus uses a stratified per-class cap of 15,000 rows at seed 42, and replaying that cap leaves 307 of the 490 gold reviews unseen, 306 of them scorable. The cap binds only on the four classes V5 predicts most often, so membership of the unseen set is decided by V5's *predicted* class rather than at random. Scored by expert label the unseen 307 still spans all seven classes, including 38 `compatibility`, 32 `usability`, and 31 `performance` reviews, so it does cover the classes carrying the recovery; what it cannot rule out is that reviews V5 predicted confidently into large classes, which may be easier, are over-represented there. We release the replay script, since the split is a faithful reconstruction rather than a recorded fact.

To rule out memorization of the 200 synthetic compatibility templates used during V5 training, we evaluate V5 against Maalej's 5,008 labels under a label-space cross-protocol test: predictions are scored against Maalej's taxonomy, which has no `compatibility` class, so any compatibility prediction is out-of-distribution with respect to the evaluation labels. On the five overlapping classes ($n = 4{,}560$), V5 achieves macro F1 = 0.676, weighted F1 = 0.730, and accuracy = 72.0%. V5 also flags 432 reviews (8.6%) as `compatibility`; the top-twenty-confidence flags read as canonical compatibility complaints ("crashes on iphone 5 and ipad 3"), all labeled `bug_report` in Maalej because Maalej offers no alternative, so on this evaluation the detector performs concept-recognition rather than template-matching.

A caveat about the class itself, found by auditing the training file. Of the 310 `compatibility` rows V5 trains on, 300 are augmentation. The corpus is lowercased and lemmatised throughout, and not one of its 215,583 rows contains a single uppercase character, whereas 194 of the 200 synthetic rows do. V5 is a cased RoBERTa, so any capital letter is a free and perfect marker for "synthetic compatibility". The internal test F1 of 0.74 for this class is therefore not a meaningful number and we withdraw it. The gold-standard compatibility F1 of 0.83 is not affected by the cue, because the gold is lowercase corpus text where it never fires; it is carried instead by the 100 mined rows, which are the verbatim duplicates described above. The cross-protocol result stands, since Maalej's text is also not from the synthetic set, but a reader should treat every compatibility number in this paper as resting on 300 augmented rows rather than on natural corpus evidence.

5 Cluster purity by issue type

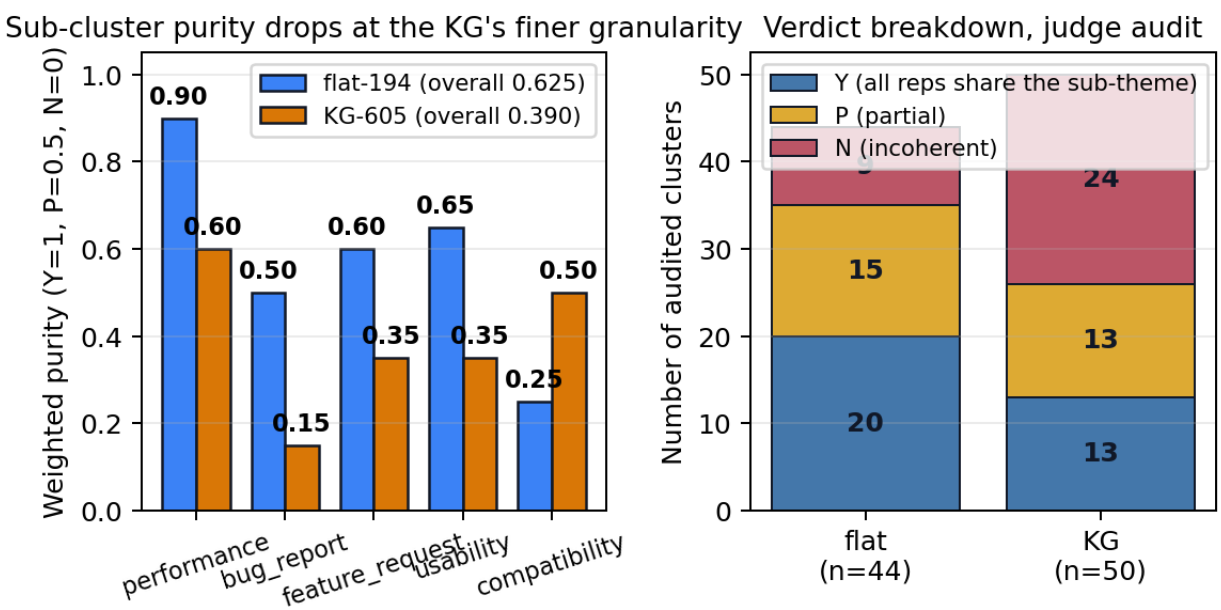

Figure 1: Extrinsic cluster purity for the flat baseline and the KG hierarchy, from the saved judge audit. *Left:* weighted purity per issue type, flat-194 against KG-605. The KG loses on four of five types, which is the finer-granularity cost we report rather than a result in its favour. *Right:* the underlying Y/P/N verdict counts behind the two overall numbers, 0.625 for flat and 0.390 for the KG. This figure is the evidence for the A2 ablation and for the claim in Section **??** that the KG's contribution is prioritization, not cluster quality.

6 Stage-5 alignment, full results

This is the analysis Section 5 summarises.

Five policies were trained on distilGPT2 (settings in the supplementary material). Head-to-head on a 100-review test set, BLEU-1 / ROUGE-L / BERTScore F1 are 0.090 / 0.097 / 0.802 for SFT-base, 0.068 / 0.074 / 0.792 for KTO, 0.084 / 0.088 / 0.800 for DPO, **0.137 / 0.131 / 0.806** for the constrained proxy, and 0.087 / 0.091 / 0.798 for Lagrangian PPO, so the constrained proxy leads three of four metrics with +53% BLEU-1 over SFT-base; the rule-based safety-violation rate is uninformative here because compliance is satisfied at initialisation.

The dual-objective claim is not demonstrated, and we report that as the result. At distilGPT2 scale the compliance scorer is satisfied at initialisation ($C = 0.94 \geq \tau$), so λ collapses to zero and the update reduces to single-objective optimisation. Two further experiments show this is a property of the regime rather than of one threshold choice: a re-run with the four operational compliance checks and τ tightened from 0.50 to 0.90 records 0 binding steps out of 30 and one violation across the run, and a sweep over $\tau \in \{0.0, 0.5, 0.7, 0.9, 1.0\}$ leaves the pass rate at 100% for four of the five policies at every threshold, only SFT-base dipping to 99% at $\tau \geq 0.9$. A model of this size cannot plausibly violate the rubric, so the CMDP machinery is untested under binding conditions. A second explanation we cannot separate from the first weakens the finding further rather than rescuing it. The operational scorer is a regular expression over roughly thirty tokens, and we tested its sensitivity by hand: nine replies that plainly breach the four rubric dimensions but are phrased away from the patterns ("Our engineers have committed to a permanent resolution by Friday") all score a perfect 1.00, and empty output, whitespace, and gibberish score 1.00 too, so a policy that collapses to emitting nothing is recorded as perfectly compliant. The quality scorer has the mirror defect: eleven rubric keywords in a bare list score 1.000 against 0.672 for a genuine reply. Our sweep therefore establishes only that these policies rarely emit about thirty specific tokens; whether they violate the four behaviours the rubric names this instrument cannot tell us, and we no longer claim it does. Every compliance number here is a property of the detector, and the +53% BLEU-1 figure is a single-objective result, not evidence of a quality-versus-compliance trade-off.

The constraint binds once the policy can violate it, as this instrument measures violation. We originally read the obstacle as capability rather than parameter count: distilGPT2 never produces the fluent phrasing the rubric penalises, so it cannot violate. Given the detector's insensitivity that reading is not the only one, and every violation rate below carries that ambiguity. We first justified it by observing that 5.2% of our own Stage-4 generations trip the scorer. That figure is from the four withdrawn template-composed arms (21 violations in 400 replies); scored the same way, the two model-generated arms we actually defend violate on 4 of 200 replies, 2.0%, one of them an information leak rather than an over-promise. The zero-shot Qwen2.5-1.5B probe gives 2 violations in 40 replies, 5.0%, whose 95% interval of roughly 1–17% cannot be distinguished from 2.0% either way, and distilGPT2 gives roughly zero. The honest version is weaker than the one we first made: a 1.5B instruct model violates occasionally where distilGPT2 essentially never does, enough for the constraint to bind, but we cannot claim it matches the rate of our own generations. Re-running the Lagrangian loop with Qwen2.5-1.5B-Instruct as the policy (the supplementary material) inverts the behaviour: over 90 steps the constraint binds on 13 (14.4%), and λ leaves zero, rises monotonically to 0.65 (0.05, then 0.40, then 0.65 by thirds), and never collapses back, while a 30-step run reaches only $\lambda = 0.125$, so the multiplier scales with the horizon. The claim that the CMDP degenerates to single-objective optimisation is therefore a property of the proof-of-concept policy, not of the formulation.

What is and is not shown. Quality moves from 0.565 to 0.590 and compliance from 0.997 to 0.989 while λ climbs, but we do not call this a trade-off: the step-level correlation between the two is $\rho = -0.12$ ($p = 0.25$), two separate trends over one run with no measured coupling. Nor can we show the multiplier growing large enough to pull the policy back into compliance; at a 5% base violation rate and a batch of four, roughly one violating sample arrives every seven steps, and 90 LoRA steps do not accumulate enough pressure to reverse the drift. So the dual objective engages and the multiplier responds monotonically to the tokens it matches, the tension between objectives is not measurable on our data, and closing the loop needs a longer horizon or a higher base violation rate than a 1.5B instruct model supplies. What this layer contributes is the formulation instantiated over four dev-rel constraints, a trainer that runs, the threshold sweep, the capability probe, the binding-constraint re-run, and the released testbed.

7 Cluster quality, full results

Stage 2 was redesigned from a flat UMAP+HDBSCAN baseline (194 clusters, mean size 471.0) to the three-layer KG-hierarchical design (605 sub-clusters, mean size 15.9), but the two were not run over the same reviews: the flat baseline clusters 91,368 reviews from the full relabelled corpus, whereas the KG design is built on the 8,404-review sample, of which 5,689 land in a sub-cluster. That sixteen-fold difference is a third reason, on top of the two below, why the head-to-head intrinsic comparison is not a comparison. We compute intrinsic geometry (Davies-Bouldin [**?**], Calinski-Harabasz [**?**], silhouette [**?**], following [**?**, **?**, **?**]) and extrinsic purity [**?**] under the Y/P/N weighted rubric of the supplementary material.

We reported that KG-hierarchical achieves 5.4× lower Davies-Bouldin and 1.9× higher Calinski-Harabasz than the flat baseline. We withdraw those two figures. The script that produced the flat baseline's intrinsic metrics draws its cluster labels as a random subsample of 10,000 reviews spanning all 194 clusters, but collects the review texts to embed by walking the clusters in order, and the first flat cluster alone contains 10,507 reviews. The embeddings therefore come from one cluster while the labels span 194, so the flat DB of 12.15 and CH of 0.98 measure random labelling rather than the flat partition, and the comparison was a valid measurement (the KG's DB 2.24 and CH 1.85, on a correctly paired set) against a broken one. Repairing the alignment would not make the two comparable anyway, since the flat side embeds 10,000 full reviews and the KG side three representatives per cluster. The count-matched A1b ablation, which embeds representatives on both sides, is the only intrinsic comparison we report, and it is unaffected. Extrinsic purity is also unaffected: the KG's finer clusters score 0.39 against the flat baseline's 0.625 under the same Qwen judge, and the flat baseline's 0.66 lead-author purity sits in the substantial-agreement band (≥ 0.61) against KG-605's fair band (≥ 0.21) on the Landis-Koch [**?**] convention adapted to weighted purity, reflecting the granularity-versus-coherence trade-off A1b makes explicit. Per issue type the KG loses on four of five.

Where the KG actually wins. On the prioritisation axis the story reverses: of the 10,534 aspect nodes, the top five by PageRank are touched by 55% of reviews and the top ten by 66%, so a developer triaging in that order reaches 55% of the complaint volume after five aspect buckets. These are the deduplicated figures; counting a review once per aspect it mentions inflates them to 74% and 96%, and that pair should not be used. We previously wrote "three-quarters" here, the discarded 74% reappearing two lines after we told the reader not to use it, and also claimed a "roughly $10^3\times$ concentration ratio" against random aspect ordering. We withdraw that too: we never computed it, and random ordering is the wrong baseline anyway, since the operative comparison is ordering by raw frequency, which we did not run. The KG additionally enables per-aspect drill-down, distinguishing `battery`→Samsung-drain from `battery`→fast-charge.